

Retrieval Algorithms — Basic, MMR, and HYDE

Lesson 04 · Studio: Grounding Agents in Knowledge · Lyzr University

What you'll learn

- Understanding vector similarity match scores and decimal thresholds
- When to use Basic Cosine Similarity vs. Maximal Marginal Relevance (MMR)
- How HYDE (Hypothetical Document Embeddings) resolves vague user queries
- Tuning chunk limits and score cutoffs for high-precision retrieval

Key Concepts

Basic Similarity

Computes vector distance between query and stored chunks. Fast and accurate for direct queries.

Maximal Marginal Relevance (MMR)

Balances relevance with diversity. Prevents returning duplicate chunks when documents have repetitive text.

HYDE (Hypothetical Embeddings)

Generates a hypothetical answer first, then retrieves chunks matching the hypothetical answer vector. Ideal for generic or short queries.

Code & Workflow Reference

```
# Configuring advanced retrieval algorithms in Studio
kb.configure_retrieval(
    type="HYDE",           # Options: "basic", "MMR", "HYDE"
    top_k=4,               # Number of chunks to retrieve
    score_threshold=0.60   # Minimum similarity cutoff (60%)
)
```